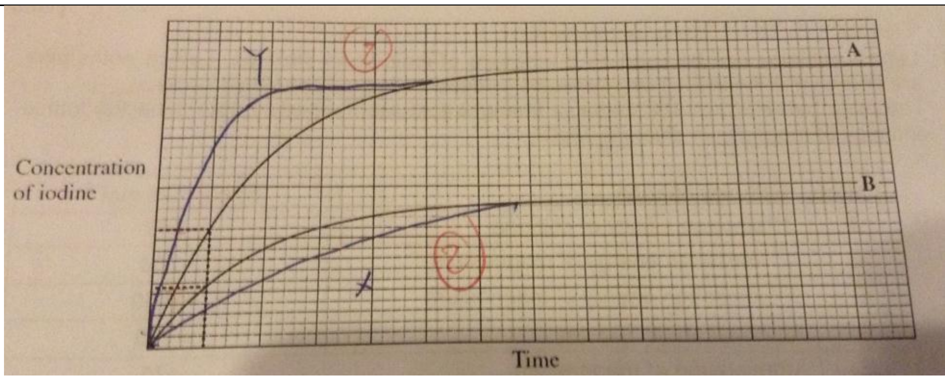
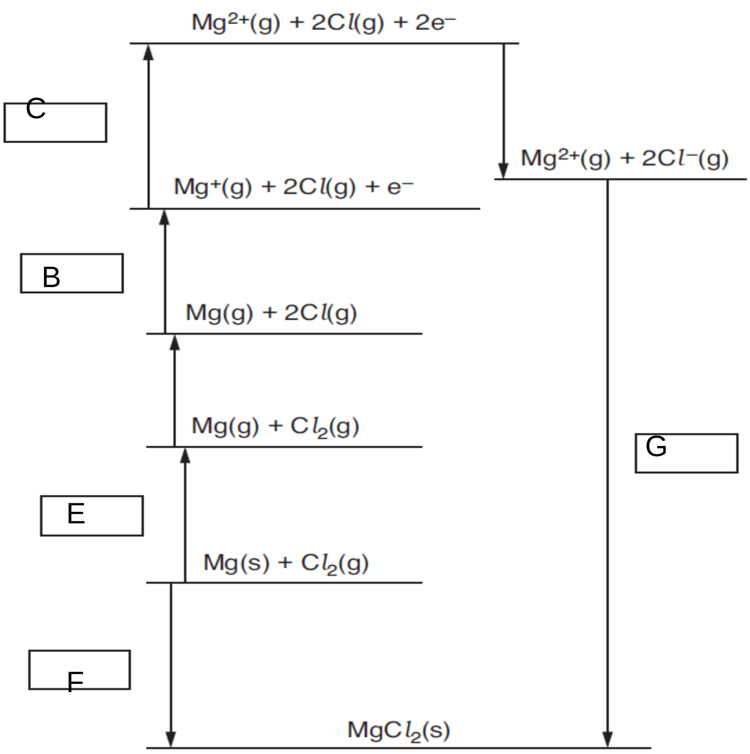
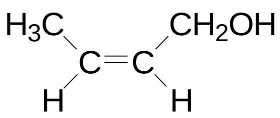
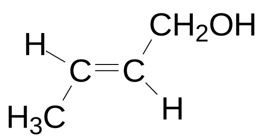
1(a)	Cr^{3+}: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$ Mn^{2+}: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$		
(b)	(i)	CH_3CH_2OH $CH_3CH(OH)CH_3$	
	(ii)	I: Orange to green II: No. of moles of $K_2Cr_2O_7 = \frac{47.2}{39.1\times2+52\times2+16\times7}$ $= 0.160$ III: No. of moles of $C_3H_6O = \frac{18}{60}$ $=0.3$ C_3H_6O is the limiting reagent. Theoretical mass of C_3H_6O produced = 0.3 x (36+6+16) = 17.4g % yield = $\frac{5.22}{17.4} \times 100\%$ = 30%	
	(iii)	I: Impurity – carboxylic acid Reason – Chemical shift of 9.0-13.0ppm represents the presence Of R-COOH group. II: CH_3CH_2OH	

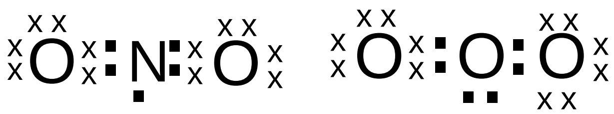
2(a)	Order: 1^{st} order Explanation: The same amount of time taken gives twice the concentration for curve A when compared to curve B.	
(b) (c)		
(d)	Reaction between I^- and $S_2O_8^{2-}$ has high activation energy as they are of the same charge and energy is required to overcome the repulsive forces.	

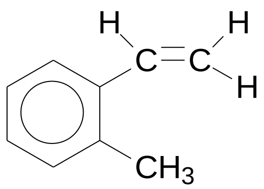
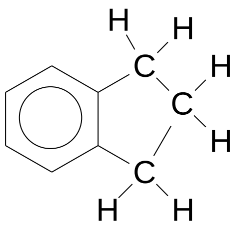
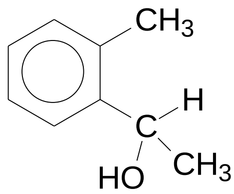
3(a)		

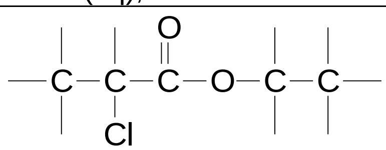
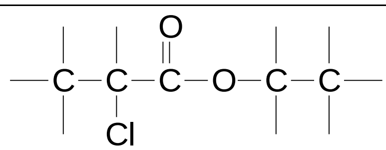
(b)	(1450+736+2x150+76+642) – (2x349) = -2506 kJmol^{-1}	
(c)	$\lvert L.E \rvert \propto \frac{q^{+} \times q^{-}}{r^{+}+r^{-}}$ Since $r_{Cl^-} > r_{F^-}$, therefore the lattice energy of magnesium fluoride is more exothermic than magnesium chloride. Hence more energy is required to break the ionic bond between magnesium ion and fluoride ion.	

4(a)	C_4H_8O		
(b)	(i)	Test 3: alkene	
	(ii)	Test 4: alcohol	
(c)	(i)	Carbonyl	
	(ii)	1° alcohol	
(d)		Free rotation about the bond is restricted by the presence of multiple bonds. The carbon atoms on both ends of the C=C double bond contain two non-identical groups.	
(e)		 Cis  trans	

5(a)	(i)	-2	
	(ii)	**A**: HCHO **B**: HCOOH **C**: CO_2	
	(iii)	**A:** $K_2Cr_2O_7$, dilute sulfuric acid and heat with distillation **B:** $K_2Cr_2O_7$ or $KMnO_4$ dilute sulfuric acid and heat **C:** Heat with excess hydrogen or $KMnO_4$ dilute sulfuric acid and heat	
(b)	(i)	$Q_{50\%}$= 100 x 4.18 x 30 = 12 540J $n_{ethanol}$ = 1.5/46 = 0.0326 moles $\Delta H=-Q_{100\%}/n_{ethanol}$ = -769 $kJmol^{-1}$	
	(ii)	There will be heat lost to the surrounding.	
(c)	(i)	$K_c = [CH_3OH]/[H_2]^2[CO]$ $mol^{-2}dm^6$	
	(ii)	$K_c = (0.8/1.5) / (0.4/1.5)^2(0/4/1.5)$ = 56,25	
	(iii)	[2(436)+1077]-[3(410)+360+460] ΔH = -101 $kJmol^{-1}$	
(d)	(i) (ii)	D: $CH_3CH(CH_3)CH_2OH$ Primary E: $CH_3CH_2CH_2CH_2OH$ Primary F: $CH_3CH_2CH(OH)CH_3$ Secondary G: $(CH_3)_3COH$ Tertiary	
	(iii)	F	
	(iv)	F	

6(a)	The total number of protons and neutron in the nucleus		
(b)	Deflection to the positive plate The angle of deflection is greater.		
(c)	(i)		
	(ii)	Bond angle of O_3 is between 110° to 120° because lone pair-bond pair replusion is greater than bond pair-bond pair replusion. The bond angle of NO_2 is between the bond angle of O_3 to 170° as a single unpaired electron will be less effective in diminishing the basic angle, and so the bond angle should be greater than that for O_3.	

	(iii)	Chlorine is able to expand its octet of electrons through the 3d orbital which is accessible for electron occuptation. Fluorine cannot expand its octer and is unable to form dative bond as it is too electronegative.	
(d)	(i)	For magnesium to react with oxygen it is necessary to apply heat, ideally by holding a piece of the metal in tongs in an atmosphere of oxygen, such as in a gas jar. Magnesium burns on heating with a brilliant white flame.	
	(ii)	NaCl: dissolves but do not under go hydrolysis $MgCl_2$: undergoes partial hydrolysis $AlCl_3$: Undergoes hydrolysis $SiCl_4$: Undergoes hydrolysis and release HCl PCl_5: Undergoes hydrolysis and release HCl	
(e)	(i)	No. of moles of $S_2O_3^{2-}$ = 0.0015 No. of moles of I_2 = 0.00075	
	(ii)	No. of moles of O_3 = 0.00075 Volume of O_3 = 16.8 cm^3 % of O_3 present = 3.36%	
(f)	(i)	Aldehyde: $K_2Cr_2O_7$, dilute H_2SO_4, heat with immediate distillation Carboxylix acid: $K_2Cr_2O_7$/$KMnO_4$ dilute H_2SO_4, heat	
	(ii)	 **L**  **M**  **N**	

7(a)	(i)	NaCN, dilute H_2SO_4, trace amount of NaCN	
	(ii)	Dilute H_2SO_4, heat	
(b)	(i)	Assuming lactic acid is a strong acid, pH = -lg[H^+] = 0.699 Since the pH of lactic acid is 1.7, it will be a weak acid as it is only partially dissociated.	
(c)	(i)	It is a solution that will resist pH changes when small amount of acid or alkali is added. $CH_3CH(OH)COOH + OH^- \rightarrow CH_3CH(OH)COO^- + H_2O$ $CH_3CH(OH)COO^- + H^+ \rightarrow CH_3CH(OH)COOH$	
	(ii)	Phenolphthalein, colourless to pink	
(d)	(i)	Ester	
	(ii)	Hydrolysis	
(e)	(i)	NaOH(aq), heat	
	(ii)	 **K**  **L**	

		The COCl is more reactive than C-Cl due to extra electronegative oxygen in COCl will polarise the C-Cl bond to a greater extent which make it easier to break and hence more reactive.	
(f)	1) Test: $Na_2CO_3(aq)$ Observation: N – effervescence M – no effervescence 2) Test: 2, 4-dinitrophenylhydrazine Observation: N – no orange precipitate M – orange precipitate 3) Test: $[Ag(NH_3)_2]^+$ Observation: N – no silver mirror M – silver mirror 4) Test: Alkaline solution of copper(II) tartarate Observation: N – Blue solution M – reddish brown precipiate		